
Algorithm 3: Stabilizing $\mathcal{W}$ -games

Input: N players, $\mathcal{S} = \{(S_j, \vec{v}(S_j))\}_{j=1}^m, \varepsilon > 0$ **Output:** A partition π

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1  $\vec{v}^* \leftarrow \text{LEARN-}\mathcal{W}\text{-GAMES}(N, \mathcal{S})$ 
2  $\pi \leftarrow \emptyset$ 
3 while  $N \neq \emptyset$  do
4   Pick  $i \in N$ 
5   if  $N \setminus \{i\} \neq \emptyset$  then
6      $j \leftarrow \arg\max_{k \in N \setminus \{i\}} v_i^*(k)$ 
7      $\pi \leftarrow \pi \cup \{\{i, j\}\}, N \leftarrow N \setminus \{i, j\}$ 
8   else  $\pi \leftarrow \pi \cup \{\{i\}\}$ 
9 return  $\pi$ 

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Proposition 5.8. Let $\varepsilon, \delta > 0$ and $\mathcal{S}$ be a sample of size m . If $m \geq \frac{2\lambda}{\varepsilon} \log \frac{n^2}{\delta}$, LEARN- $\mathcal{W}$ -GAMES returns an ε -estimate $\vec{v}^*$ of $\vec{v}$ with confidence $1 - \delta$.

We want to show that, by relying on the ε -estimate given by LEARN- $\mathcal{W}$ -GAMES for ε “not too small”, Algorithm 3 returns an ε -stable partition. We first state a technical lemma.

Lemma 5.9. Let π be the output of Algorithm 3 and let us call a player i green if it is not in a coalition with one of his $\lfloor \log_2 \frac{1}{\varepsilon} \rfloor$ least preferred choices according to v_i^* . Then,

- a) for i green, $\Pr_{S \sim \mathcal{D}} [i \in S \wedge v_i(S) > v_i(\pi(i))] < \lambda\varepsilon$,
- b) for $\varepsilon \geq \sqrt[3]{\frac{\lambda^2}{2n}}$, $\Pr_{S \sim \mathcal{D}} [S \text{ does not contain green } i] < \varepsilon$.

We are now finally ready to prove the main theorem, stated in the beginning of this subsection.

Proof of Theorem 5.5. For $\delta > 0$ and $\varepsilon \geq \sqrt[3]{\frac{\lambda^5}{2n-3}}$, we call Algorithm 3 with $\varepsilon' = \varepsilon/2\lambda$ and $m \geq \frac{2\lambda}{\varepsilon'} \log \frac{n^2}{\delta} = \frac{1}{\varepsilon} \log \frac{n^2}{\delta}$, and obtain an ε -stable partition with probability at least $1 - \delta$. By point a) of Lemma 5.9, a green node has probability $< \lambda\varepsilon' = \varepsilon/2$ to get a better outcome by moving from $\pi(i)$ to S . Furthermore, ε' satisfies the requirement of point b) in Lemma 5.9, so the probability of sampling an S without a green node is $< \varepsilon' \leq \varepsilon/2$. In conclusion, if we call G the event that S contains a green player, then since $\Pr_{S \sim \mathcal{D}} [S \text{ core blocks } \pi]$ equals

$$\Pr_{S \sim \mathcal{D}} [S \text{ core blocks } \pi \mid G] + \Pr_{S \sim \mathcal{D}} [S \text{ core blocks } \pi \mid \bar{G}],$$

we see that $\Pr_{S \sim \mathcal{D}} [S \text{ core blocks } \pi] \leq \varepsilon/2 + \varepsilon/2 = \varepsilon$.

For $\delta > 0$ and $\varepsilon < \sqrt[3]{\frac{\lambda^5}{2n-3}}$, with a sample of size $m \geq \frac{8\lambda^6}{\varepsilon^3} \log \frac{n^2}{\delta}$, we can reveal the exact valuation functions with probability at least $1 - \delta$ and return the core stable partition π using the algorithm of ?. Indeed, since the probability of drawing any coalition is by Equation (2) at least $1/\lambda 2^n$, this also holds for the coalition containing only agents i and j , which provides both $v_i(j)$ and $v_j(i)$. The probability of not drawing a particular coalition of size 2 is $\leq (1 - 1/\lambda 2^n)^m \leq e^{-m/\lambda 2^n} \leq \delta/n^2$. Taking a union bound over all the $\binom{n}{2} < n^2$ coalitions of size 2, we see that the probability of not seeing all the exact valuations is upper bounded by δ . $\square$

5.2 Barriers to the Restricted Distributions Approach

Encouraged by the positive results of the last subsection, one could try to extend the approach of focusing on bounded distributions in the hope that other classes that are known not to be PAC stabilizable, such as Additively Separable, Fractional, and Anonymous HGs, are in fact stabilizable under such distributions. Unfortunately, this does not seem to be always the case, as we discuss below.

Definition 5.10. For a HGs class $\mathcal{H}$, let $T(\mathcal{H})$ be the time complexity of the best algorithm solving the core for this class, i.e., the runtime of the fastest algorithm that for every input instance either correctly replies that the core is empty or returns a core-stable partition.

Theorem 5.11. If $T(\mathcal{H}) = \omega(\text{poly}(2^n))$ for a HG class $\mathcal{H}$, then $\mathcal{H}$ is not efficiently PAC stabilizable, even under the uniform distribution.

Notice that the assumption $T(\mathcal{H}) \in \omega(\text{poly}(2^n))$ is not that strong. In most of the HGs classes the complexity of the problem of deciding the existence of the core is either Σ_2^P -complete or NP-hard. Although this does not imply that it is not possible to find a $O(\text{poly}(2^n))$ algorithm (in the case of the total collapse of the polynomial hierarchy, this would even be possible in polynomial time), such algorithms are currently not known and at this point it seems that finding them is unlikely. In particular, the brute force approach, that searches for an element in the core by examining all the possible partitions, has a running time of $\Omega((n/2)^{n/4})$ (as this is one possible lower bound on the Bell number), and thus its running time is also in $\omega(\text{poly}(2^n))$.

6 Conclusions

In this work, we initiated the study of a unified approach for determining the learnability and stabilizability of specific HGs classes. One of the obvious goals for future work is finding a unique characterization of HCN representations that imply learnability. Another one is exploring further consequences of Theorem 5.11 and expanding the knowledge on the exact computational complexity of solving the core for the different classes of HGs.

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